11 — Supplementary Methods, Tables, and Figures

SEION-KGR v26 MAX

Campaign ID: gate12-closeout-2026-08-01
Canonical commit: `6af3c35271ae2ffab41ecba2aad098d1988fdc0c`
Branch: `campaign/gate12-closeout`
Date: 2026-08-01

Complete hyperparameters, seed-level tables, test inventory, artifact manifest, exact commands.

Contents

1 Complete Tier 0 hyperparameters

```
Base-expert selection: dim=64 epochs=1 batch_size=1024 neg_k=64
  eval_max_queries=200 entity_block_eval=4096 seed=1 device=cuda

A0: dim=64 base_expert=tucker epochs=5 batch_size=1024 neg_k=64
  adversarial_temperature=1.0 n3_weight=0.001 lr=0.001
  eval_every=5 eval_max_queries=300 entity_block_eval=4096
  seeds={1,2} device=cuda datasets={FB15K-237(full), WN18RR(full)}

A3: as A0, plus enable_path enable_seion
  path_rank=16 path_layers=2 path_max_neighbors=16
  path_selector_mode=budgeted_bfs seion_rank=16
  epochs=3 batch_size=128 eval_max_queries=100
  seeds={1,2} device=cpu  (deviation, see preregistration.md Sec. 11)
  datasets={FB15K-237(5k subsample), WN18RR(5k subsample)}
```

2 Complete seed-level results

See `04_screening_report.pdf` §3 for the full 8-row table (test MRR, valid MRR, Hits@1, Hits@10, wall-clock, per config/dataset/seed). Reproduced in machine-readable form at `campaigns/gate12/tier0_results.json`.

3 Negative-control results (full)

Control	Metric	Value
Randomized labels	Structured-graph MRR	0.5764
Randomized labels	Randomized MRR	0.1392
Randomized labels	Degradation factor	4.14×
Leakage (3-epoch run)	Clean MRR	0.2961
Leakage (3-epoch run)	Leaky MRR	0.2961 (null effect)
Leakage (40-epoch follow-up)	Mean path gate $\sigma(\gamma_r)$	0.01799 (init: 0.01799)
Leakage (40-epoch follow-up)	Clean MRR	0.6833
Leakage (40-epoch follow-up)	Leaky MRR	0.6833 (still null)

Full JSON: `campaigns/gate12/negative_controls_results.json`.

4 Test inventory (130 tests, all passing)

File	Count
Canonical commit (66):	
test_algebra.py, test_projection.py, test_evaluator_exactness.py, test_gradients.py, test_negative_controls.py, test_seion_kgr_package.py, test_recursive_projection.py, test_reference_determinism.py	66
This campaign (64):	
test_selector.py (Phase B2)	14
test_structural_kernel.py (Phase B3)	19
test_certification.py (Phase B4)	15
test_perf_and_parity.py (Phase B5)	4
test_campaign_negative_controls.py (Phase D)	2

Note: file-level test counts above for the campaign additions sum to 54, not 64 — the remaining 10 are additional cases added to existing canonical-commit files during Phase B2/B5 work (e.g. the permutation-gauge tests). Run `python -m pytest tests/kgr -q` for the authoritative count.

5 Artifact manifest

Artifact	Path
Preregistration	`campaigns/gate12/preregistration.md`
Tier 0 config (frozen)	`campaigns/gate12/tier0_config.json`
Base-expert selection	`campaigns/gate12/tier0_base_selection_result.json`
Tier 0 results	`campaigns/gate12/tier0_results.json`
Tier 0 summary	`campaigns/gate12/tier0_SUMMARY.md`
Negative controls	`campaigns/gate12/negative_controls_results.json`
Claim matrix	`docs/SEION_KGR_CLAIM_MATRIX.md`
Assumption ledger	`docs/SEION_KGR_ASSUMPTION_LEDGER.md`
CI workflow	`.github/workflows/kgr-ci.yml`
CI run (green)	https://github.com/voidzeit/seion-math-core/actions/runs/30726372321

6 Exact reproduction commands

```
git clone https://github.com/voidzeit/seion-math-core
git checkout campaign/gate12-closeout
python -m pip install -e '.[test,research]'
python -m pytest tests/kgr -q
python seion_kgr_reference_fp64.py --self_test
python seion_kgr_v26_train.py --self_test
# Tier 0 (place FB15K-237/WN18RR under data/ first):
python seion_kgr_v26_train.py --train data/FB15K-237/train.txt \
  --valid data/FB15K-237/valid.txt --test data/FB15K-237/test.txt \
  --out_dir runs/A0_fb15k237_seed1 --dim 64 --base_expert tucker \
  --epochs 5 --batch_size 1024 --neg_k 64 --eval_every 5 \
  --eval_max_queries 300 --entity_block_eval 4096 --seed 1
```